Architectural Specification: Open-Science Ethological Data Ingestion Pipeline

1. Strategic Context and Scientific Necessity

The field of ethology currently faces a systemic "replication crisis" rooted not in a lack of theory, but in a lack of robust data engineering. Recent studies identify a 27% data-sharing bottleneck in animal behavior research; while researchers intend to share data, the absence of standardized technical infrastructure makes aggregation impossible. This specification outlines a pipeline designed to bridge this gap, moving beyond anecdotal data collection to create a structured, citable scientific resource.**Definition:** Current ethological data collection frequently utilizes "toy" applications or fragmented narrative records that lack the structural integrity required for epidemiological review. Modern science requires developers far more than it requires more theorists; there is a profound necessity for standardized infrastructure that allows for the aggregation of raw data into anonymized, structured datasets suitable for statistical correlation.To ensure scientific validity, this architecture rejects the traditional use of "Generative AI" in favor of **Deterministic Data Parsing** . In this context, determinism is defined as the guarantee of a predictable state and output for any given input. By utilizing large language models (LLMs) as strict parsers rather than creative engines, we ensure every data point adheres to a rigorous, verifiable schema. This shift is mandatory for producing datasets reliable enough for peer-reviewed research and public policy.

2. Technical Stack and Architectural Rationale

To meet strict delivery timelines while maintaining high code quality, this architecture consolidates the stack within the Google Cloud Platform (GCP). This minimizes the cognitive overhead associated with managing disparate Identity and Access Management (IAM) configurations and deployment workflows, prioritizing engineering focus on the core data engine.

Core Architectural Components

Component,Technology,Strategic Value
Core Engine,Python (Pandas/Pydantic),"Enforces strict data validation, normalization, and longitudinal tracking."
API Layer,FastAPI,"High-performance backend for secure, schema-validated data ingestion."
Database,Google Cloud Firestore,NoSQL storage providing an extensible ethogram for evolving research.
Automation,GitHub Actions (CI/CD),Ensures continuous reliability through automated unit testing (pytest).

The decision to utilize GCP over alternative providers like Microsoft Azure is based on Google's **silicon-to-model transparency** . Unlike enterprise providers that offer "wrapper" documentation focused on compliance and administrative orchestration, Google provides first-party control from their Tensor Processing Units (TPUs) to the Gemini model weights. This provides the deterministic API specifications—such as precise token limits and multimodal codec handling—required to build machine-readable scientific pipelines.

3. Structural Reliability: Schema Enforcement via Pydantic

Structural reliability is the primary gatekeeper of scientific data integrity. The pipeline must catch errors at the point of entry to prevent database corruption. By implementing **Pydantic models** , the system enforces strict data types and regex patterns for ethological incident tracking.The following model demonstrates the enforcement of standardized ethogram strings, including "neutral" states to provide a scientific baseline for behavioral variance.

```
from pydantic import BaseModel, Field, ConfigDict
from datetime import datetime
from uuid import UUID

class EthologicalIncident(BaseModel):
    model_config = ConfigDict(frozen=True)

    # Anonymized tracking for longitudinal data
    animal_id: UUID = Field(..., description="Anonymized unique
identifier for the subject")
    timestamp: datetime = Field(default_factory=datetime.utcnow)

    # Physiological tracking
    heart_rate: int = Field(..., gt=0, lt=300, description="Heart
rate in BPM")

    # Enforcing standardized ethogram via Regex
    # Includes 'neutral' as a mandatory baseline state
    behavior_type: str = Field(
        ...,
        pattern="^(barks|lunges|cowers|neutral|vocalizing|panting)$",
        description="Standardized behavior code from the 2024
Ethogram Matrix"
    )

    handler_notes: str = Field(..., max_length=1000)
```

By rejecting malformed data at the API entry point, the system prevents the ingestion of inconsistent metrics, ensuring the downstream database remains a clean, queryable resource.

4. Semantic Reliability: Deterministic AI Ingestion

A significant challenge in ethology is extracting data from "messy," unstructured narrative reports. This pipeline utilizes the Gemini API not as a text generator, but as a **schema-constrained state machine** .Utilizing Google's documentation on **Structured JSON outputs** eliminates arbitrary model hallucinations. By forcing the LLM to map narratives directly onto a Pydantic-defined JSON schema, we transform an inherently probabilistic model into a deterministic parser.

MIME Type Enforcement and Schema Mapping

To ensure semantic reliability, the Gemini API configuration must mandate response_mime_type: "application/json". The system provides the model with a JSON schema that mirrors the Python EthologicalIncident model exactly. This ensures that the model's output is always machine-readable and valid for immediate insertion into Firestore without further sanitization.

5. Data Persistence and Cloud Infrastructure

Google Cloud Firestore (NoSQL) was selected to support an **extensible ethogram** . Unlike relational databases that require rigid migrations, a NoSQL structure allows the incident matrix to evolve as new behavioral markers are identified by the scientific community.Deployment via serverless options— **Cloud Run** or **Cloud Functions** —is required for the MVP. This abstracts infrastructure management, allowing the research software engineer to focus on validation logic and code quality.

Technical Ethics in Architecture

True project ethics are a function of system architecture, not vendor marketing. This pipeline adheres to the following standards:

- **Data Sovereignty & PII Stripping:** Aggressive removal of Personally Identifiable Information (PII) occurs in the API layer before any record reaches persistent storage.
- **Open-Source Licensing:** All code is published under **MIT** or **Apache 2.0** licenses to allow for community auditing and reproduction.
- **Data Accessibility:** Provision of public, uninhibited API endpoints to prevent vendor lock-in and ensure researchers can access data regardless of their cloud provider.

6. Scientific Validation and Open-Source Governance

To be recognized as a valid contribution, the codebase must meet the standards of the **Journal of Open Source Software (JOSS)** . The repository must be functional, verifiable, and sustainable.

JOSS-Compliant Repository Structure

- **Automated Test Suite:** Comprehensive coverage using pytest to verify transformation logic.
- **Continuous Integration:** GitHub Actions workflow to execute tests on every push/pull request.
- **Core Documentation:**
- README.md: High-level overview and installation.
- CONTRIBUTING.md: Guidelines for community involvement.
- paper.md: The "Statement of Need" summarizing the ethological problem and the technical solution.

2026 Generative AI Disclosure Rule

In compliance with 2026 scientific standards, the repository must include a transparent **AI Usage Disclosure** . This documents exactly where Gemini was leveraged—specifically for the normalization of unstructured handler notes into the Pydantic matrix—ensuring that human architectural decisions remain the primary driver of the pipeline's integrity.

7. Implementation Roadmap: From MVP to Citable DOI

To establish a scientific identity as an independent researcher, the project must transition from a code repository to a permanent, citable record.

1. **Establish an ORCID iD:** Create a persistent digital identifier to link the contribution to the author's scientific identity.
2. **Apply an OSI-Approved License:** Ensure the GitHub repository includes an explicit LICENSE file (MIT/Apache 2.0).
3. **Sync with Zenodo:** Archive the repository on Zenodo to mint a permanent **Digital Object Identifier (DOI)** , making the software formally citable in academic literature.
4. **Register with OSF:** Submit the data schemas and validation models to the Open Science Framework (OSF) for visibility among veterinary behaviorists.**Conclusion:** The infrastructure *is* the scientific contribution. By applying rigorous data engineering to the field of ethology, this architecture transforms raw, fragmented observations into a trusted, standardized scientific resource.